

Customer Journey Map

6.PROJECT TESTING

Date	4 July 2026
Team ID	XXXXXX
Project Name	A Comprehensive Measure of Well Being
Maximum Marks	3 Marks

Performance & Unit Testing Report

This document reports the performance characteristics of our predictive engine and details the execution of our unit testing suites.

1. Machine Learning Model Performance

Our Linear Regression model was fitted on **1,152** annual country development records generated from UNDP panel statistics. The model evaluation metrics on an independent test partition (80-20 train-test split) are summarized below:

METRIC	COMPUTED VALUE	DESCRIPTION
Mean Absolute Error (MAE)	0.00979	Average absolute vertical distance between predicted and actual HDI values.
Mean Squared Error (MSE)	0.00015	Average squared vertical error distance, penalizing larger deviations.

METRIC	COMPUTED VALUE	DESCRIPTION
Root Mean Squared Error (RMSE)	0.01224	Square root of MSE, matching target units.
Coefficient of Determination (R ²)	0.99469 (\$99.47\%\$)	Proportion of target variance explainable by inputs.
Training R ² Score	0.99461 (\$99.46\%\$)	Fit accuracy on training data, indicating zero overfitting.

Feature Coefficients (Std Dev Change mapping)

- Life Expectancy: +0.13545
- Mean Years of Schooling: +0.04817
- GNI per Capita: +0.02936
- Expected Years of Schooling: -0.03857
- Intercept: 0.72836

2. Unit Testing Execution Suite

We run automated unit tests covering both the predictive backend and the web server routing.

Test Coverage Checklist

- [test_model.py](#) (2 tests):
 - test_synthetic_data_generation: Confirms panel columns structure, bounds check, and null validations.
 - test_model_training_outputs: Confirms scaler persistence, linear coefficients fit, and mock prediction limits.
- [test_app.py](#) (7 tests):
 - test_dashboard_route: Verifies the / dashboard responds with HTTP 200.
 - test_about_route: Verifies the /about route responds with HTTP 200.
 - test_invalid_404_handler: Asserts custom page templates are served for incorrect routes.

- `test_api_countries_endpoint`: Verifies list of countries is served in JSON array.
- `test_api_stats_endpoint`: Verifies dataset statistics are served in JSON format.
- `test_prediction_successful`: Verifies prediction executes correctly with valid arguments.
- `test_prediction_invalid_fields`: Asserts form boundaries reject out-of-range arguments returning error code 400.

Automated Test Results Command

```
python -m unittest discover -s tests
```

- **Execution Status: PASS (OK)**
- **Total Tests Run: 9**
- **Execution Time: \$0.404\text{ seconds}\$**
- **Success Rate: \$100\%\$**